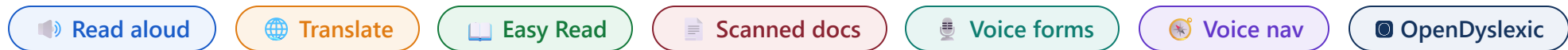




AaaS — Accessibility as a Service

One button on any government website that **reads, translates, simplifies, scans and listens** — in Odia, by voice.



Team SUBARNAREKHA · Smart Odisha Hackathon '25 · Apache-2.0

Arun, 68 — farmer near Jajpur

GOVERNMENT OF ODISHA SSEPD DEPARTMENT

No. 2214 /SSEPD Date: 12.05.2026

It is hereby notified that the monthly pension payable under the Madhu Babu Pension Yojana shall be enhanced from Rs. 500 to Rs. 1000 per month w.e.f. 01.06.2026, subject to furnishing of the requisite life certificate...

RECEIVED

- × Arthritis — he cannot type
- × Speaks only Odia — the portal is in English
- × The notice that matters is a **photograph of paper**

His government is online. Arun is not.

21 lakh

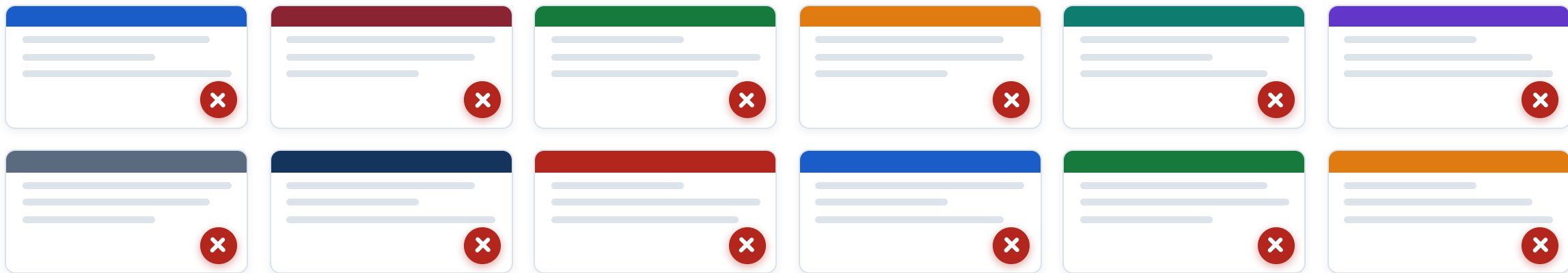
persons with disabilities in Odisha (2011, undercounted)

4.2 crore

Odia speakers worldwide

Why this stays broken

- Hundreds of portals — each fixes accessibility **alone**, badly or not at all
- Overlay widgets tick compliance boxes; they don't help a blind Odia speaker
- Screen readers can't read a scan — and read **English** at Odia speakers



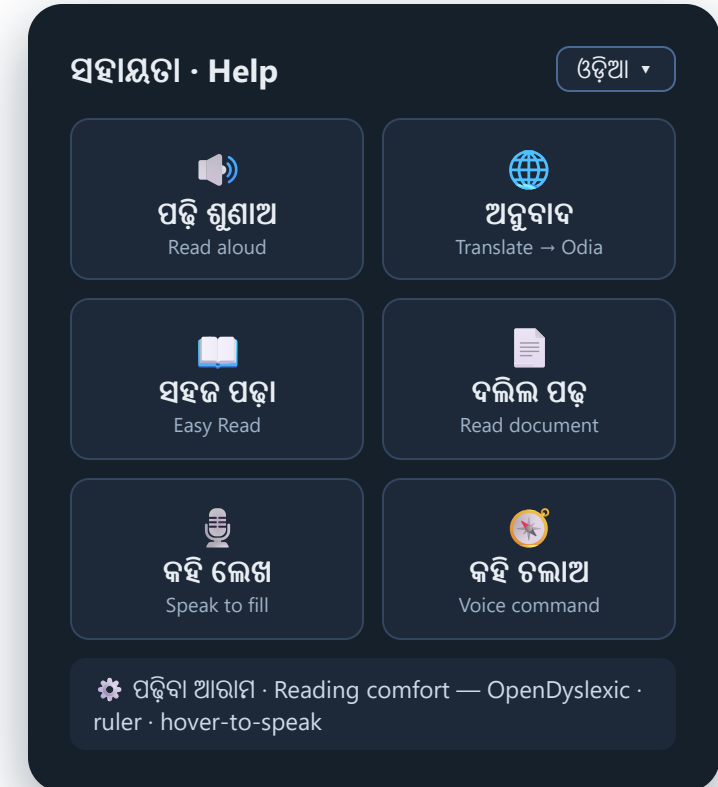
Accessibility must be **infrastructure**, not remediation



One button. Six capabilities.

-  **Read aloud** — follow-along highlight, skip / replay
-  **Translate** → **Odia** — whole page, in place, one-tap undo
-  **Easy Read** — officialese → plain language, offline rules
-  **Read documents** — OCR scans & PDFs, original + Odia **side by side**
-  **Speak to fill** — asks each field aloud; never presses Submit
-  **Voice command** — say a link's name; no dead ends

Integration = **one script tag**. Sites not integrated? **Our browser extension injects it anyway.**



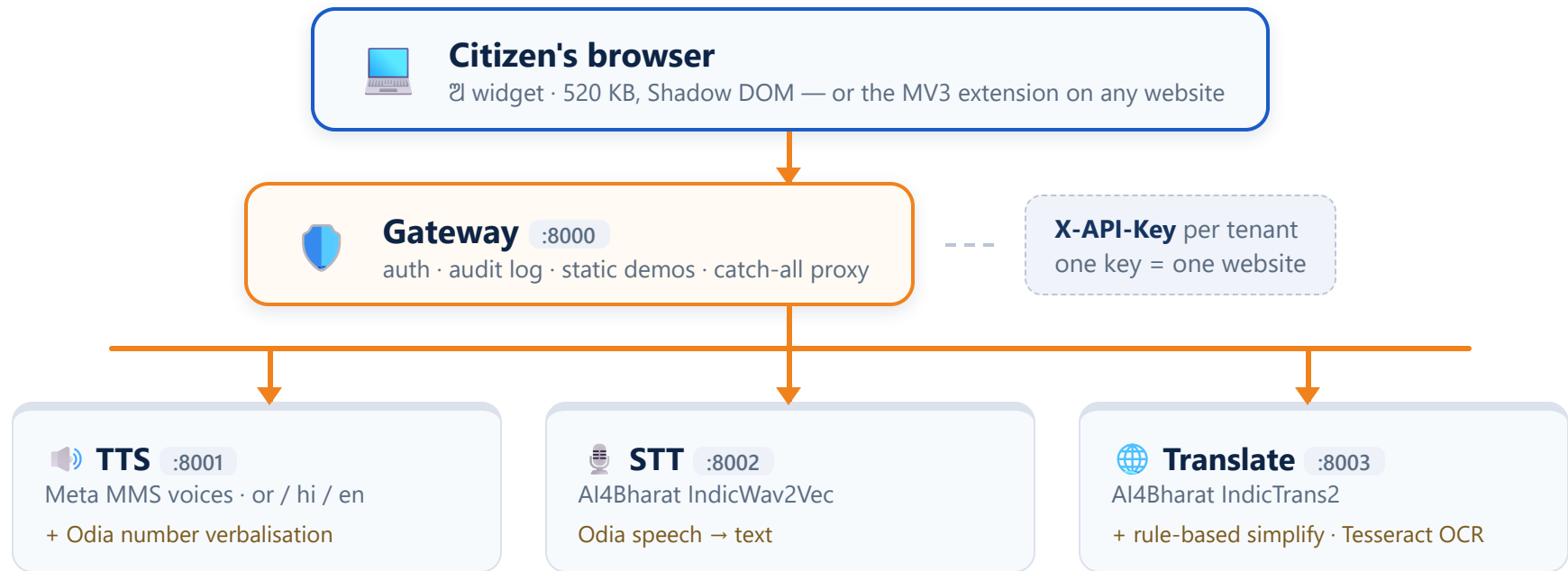
Watch Arun use his government

- 1 **Read aloud** — the SSEPD pension page speaks Odia, highlighting each line
- 2 **Translate + Easy Read** — English officialese becomes plain Odia (and back, ଓ)
- 3 **The scanned circular** — OCR → side-by-side original + Odia, read aloud
- 4 **The pension form, by voice** — name transliterated, "ଶାଠିଏ ପାଞ୍ଚ" → 65, never auto-submits
- 5 **"ବେଣ୍ଟର"** — voice navigation on the **real** jajpur.odisha.gov.in

Everything on this laptop — Wi-Fi off. ~4 minutes.



How a request flows



On-prem — OCAC in production, one laptop today · no cloud calls · every engine has an honest mock fallback · DPDP-friendly: voice processed in memory, discarded

The models are commodities. This is the product:

MMS-TTS IS MUTE ON DIGITS

"୩୦୦ ଟଙ୍କା" → **12 KB of silence**



WE VERBALISE NUMBERS

spoken as "ତିନି ଶହ ଟଙ୍କା" in Odia words

ODIA STT HEARS ENGLISH PHONETICALLY

"Contact" → "କଣ୍ଟାକ୍ଟ" — no match



ROMANISE + SOUND-MATCH

"କଣ୍ଟାକ୍ଟ" **finds and opens Contact**

TRANSLATION DESTROYS NAMES

ପୂର୍ଣ୍ଣଚନ୍ଦ୍ର → "**full moon**" in the name box



TRANSLITERATE, NEVER TRANSLATE

lands as "**Purnnachandra**"

REAL SITES HIDE THE NOTICES

tenders live inside **hidden carousels**



VOICE NAV REACHES THEM

hidden links ranked, **duplicates collapse**



It's real — six kinds of institution, one gateway

Jajpur Collectorate
district · grievance form · tender scans

Utkal University
admissions · helpdesk voice form

BSE Odisha
exam board · scanned circulars

National Scholarship Portal
central govt · "India next"

AIIMS Bhubaneswar
hospital · OPD booking by voice

SSEPD Department
flagship · every feature on one page

228
automated checks, five suites

6
real gov sites verified via extension

0
cloud calls — fully on-prem

1.7 GB
offline USB bundle, real OCR inside

Plus a live operator console: request counters that move during the demo, one-click tenant onboarding that mints a working key, and an API playground judges can press.



"Doesn't Bhashini already do this?"



AaaS — the accessibility PRODUCT

guided form-filling · name transliteration · side-by-side scanned reading · never-auto-submit safety · no-dead-end voice nav

Language ENGINES — Bhashini · AI4Bharat · Meta

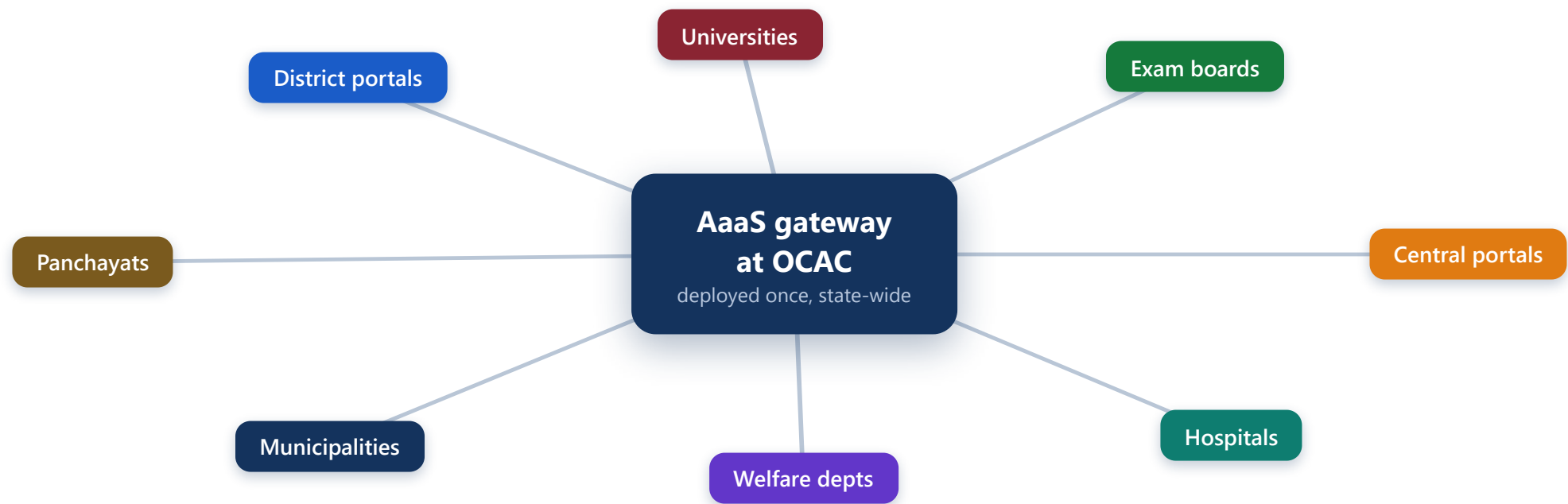
ASR · translation · TTS, as APIs — we'd plug Bhashini in tomorrow; it makes us stronger

Bhashini gives India voices.

AaaS gives Odisha's disabled citizens hands, eyes and ears.



Adoption: the UPI pattern



```
<script src="https://aaas.ocac.gov.in/widget.js" data-key="aaas_live_..." defer></script>
```

- One script tag per site
- Apache-2.0 — no lock-in
- Any SI can operate it
- Other states adopt free



Honest today · concrete next

Limits we'll tell you first

- OCR: printed notices yes — handwriting no
- STT wants clear speech — so we retry, and show tappable choices instead of guessing
- The USB bundle ships mock translation (torch doesn't fit); the laptop runs real IndicTrans2

Roadmap — no rewrites, just pilots

- Pilot: one district portal + one board site
- Native-speaker polish of the Odia glossary
- Bhashini as an alternate engine backend
- More languages — the engines already support them; it's configuration



Odisha first. India next.

Arun shouldn't need his grandson to talk to his own government.

One pilot is all we ask.

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